Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:** Design Butterworth and Chebyshev filter using bilinear transformation method**.**	
Exp:- 4	**Date: 30/09/2026**	**Enrollment No: 92400133074**

Github - https://github.com/pateldhananjay39/DSIP.git

AIM: Design Butterworth and Chebyshev filter using bilinear transformation method**.**

Code:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import butter, bilinear, freqz, cheby1

def design_butterworth_filter(filter_order, cutoff_frequency, sampling_frequency):
    analog_b, analog_a = butter(filter_order, cutoff_frequency, analog=True,
btype='low')
    digital_b, digital_a = bilinear(analog_b, analog_a, sampling_frequency)
    return digital_b, digital_a

def design_chebyshev_filter(filter_order, cutoff_frequency, sampling_frequency,
ripple):
    analog_b, analog_a = cheby1(filter_order, ripple, cutoff_frequency,
analog=True, btype='low')
    digital_b, digital_a = bilinear(analog_b, analog_a, sampling_frequency)
    return digital_b, digital_a

def plot_filter_response(digital_b, digital_a, sampling_frequency):
    frequency, magnitude_response = freqz(digital_b, digital_a,
fs=sampling_frequency)

    plt.figure(figsize=(10, 6))
    plt.plot(frequency, np.abs(magnitude_response))
    plt.title('Filter Magnitude Response')
    plt.xlabel('Frequency (Hz)')
    plt.ylabel('Magnitude')
    plt.grid(True)
```

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```
plt.show()
```

```
    _, impulse_response = freqz(digital_b, digital_a, fs=sampling_frequency,
worN=4096)

    plt.figure(figsize=(10, 6))
    plt.plot(impulse_response)
    plt.title('Filter Impulse Response')
    plt.xlabel('Samples')
    plt.ylabel('Amplitude')
    plt.grid(True)
    plt.show()

filter_order = 4
cutoff_frequency = 1000
sampling_frequency = 8000
ripple = 0.5

digital_b, digital_a = design_butterworth_filter(
    filter_order,
    cutoff_frequency,
    sampling_frequency
)

plot_filter_response(digital_b, digital_a, sampling_frequency)

digital_b, digital_a = design_chebyshev_filter(
    filter_order,
    cutoff_frequency,
    sampling_frequency,
    ripple
)

plot_filter_response(digital_b, digital_a, sampling_frequency)
```

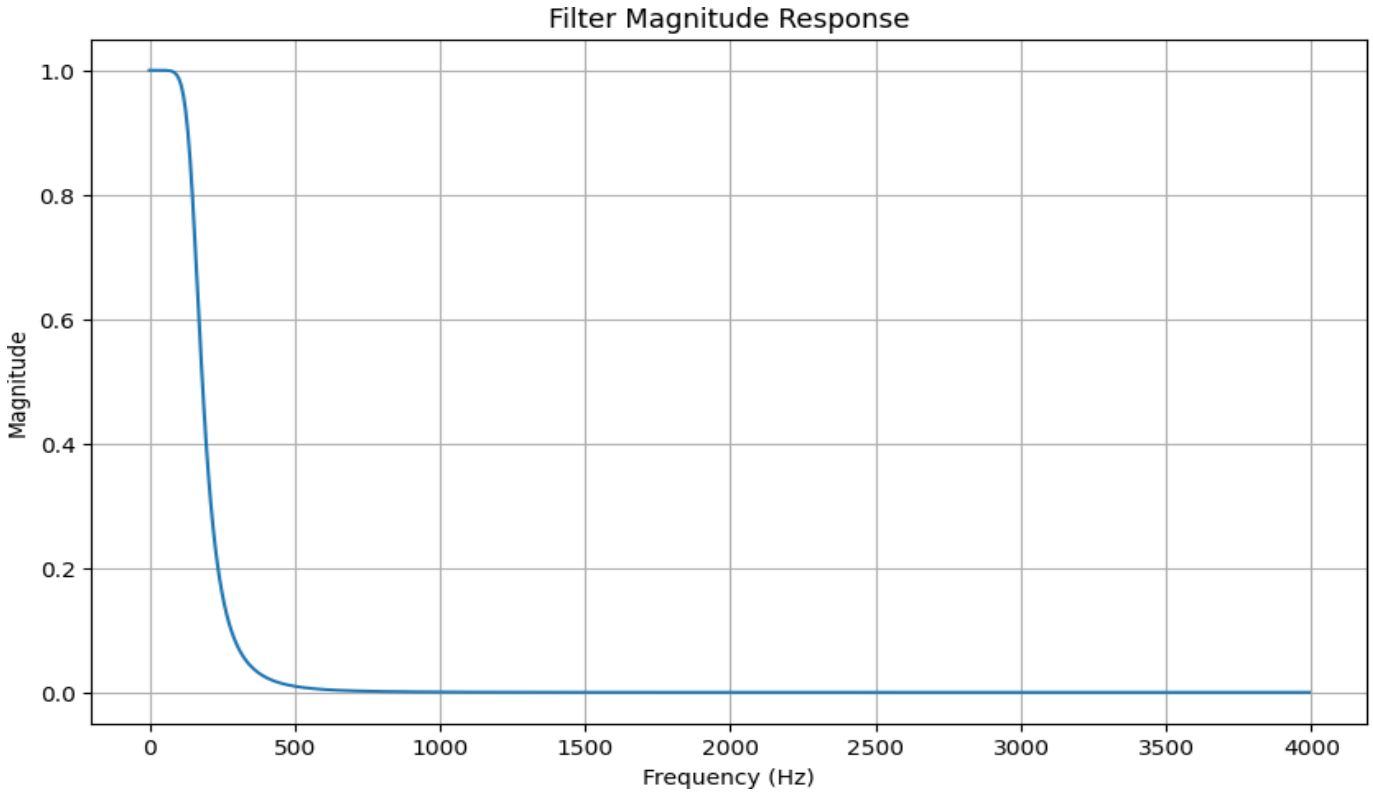
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```
filter_path = 'filter_coefficients.txt'

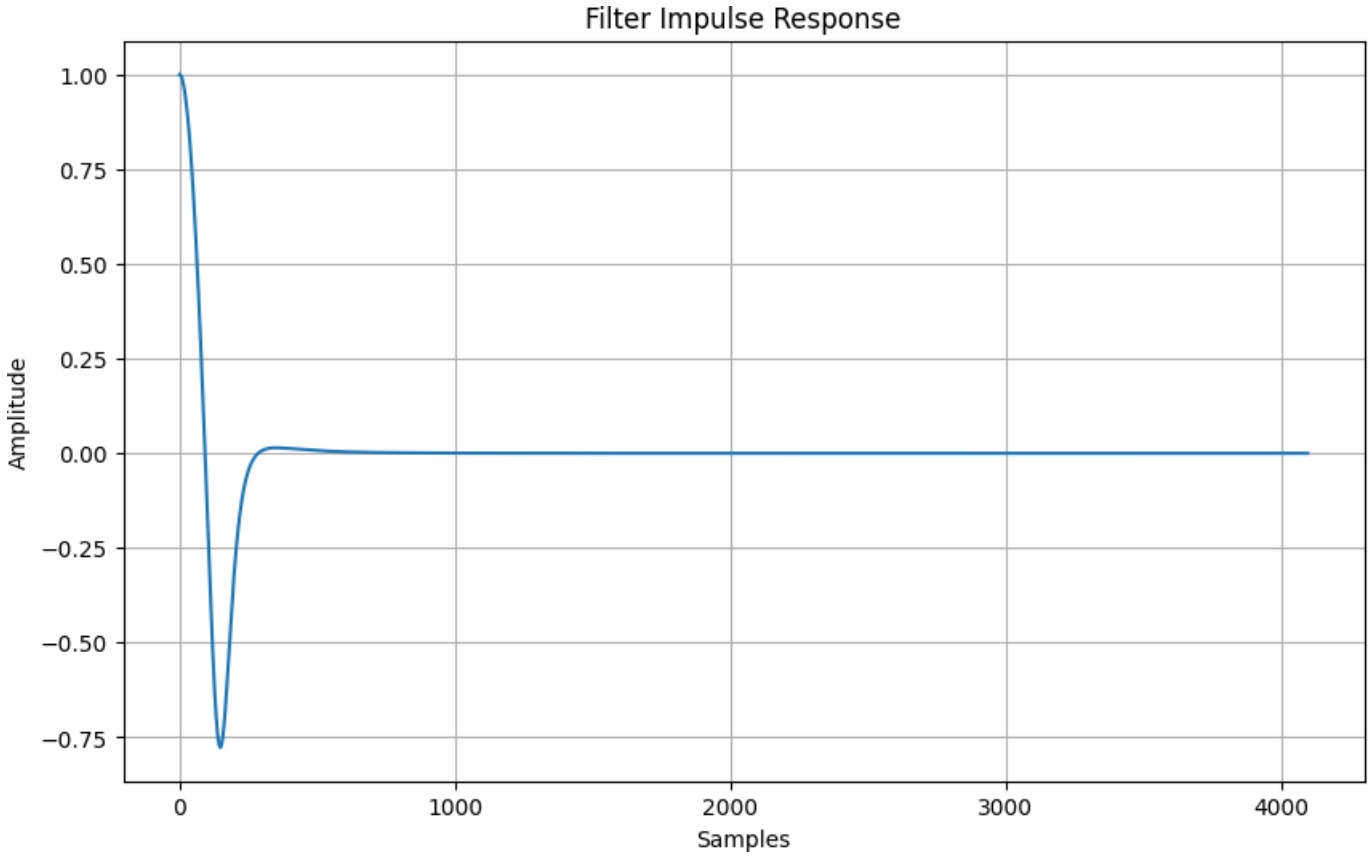
np.savetxt(
    filter_path,
    np.vstack((digital_b, digital_a)),
    delimiter=','
)

print(f"Filter coefficients saved at: {filter_path}")
```

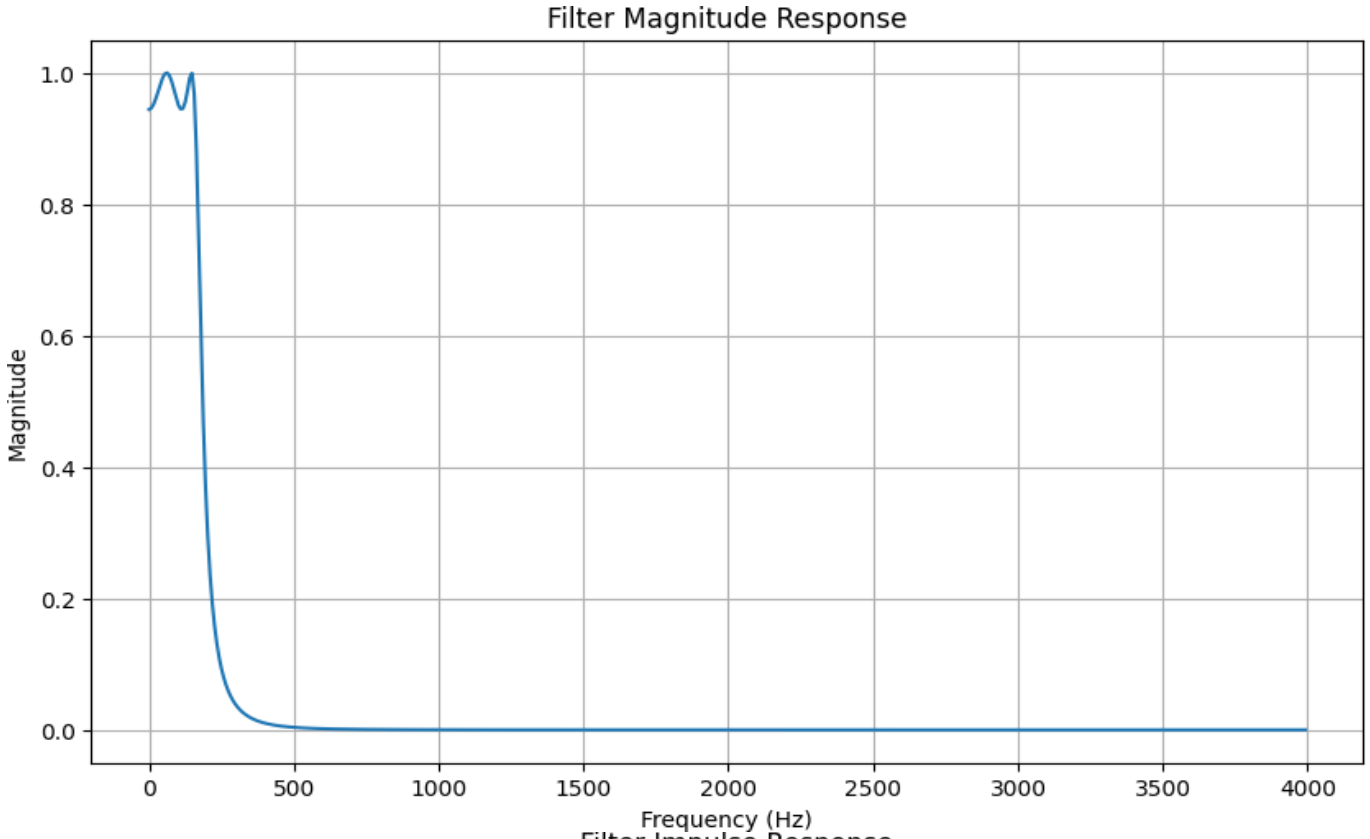
Output:

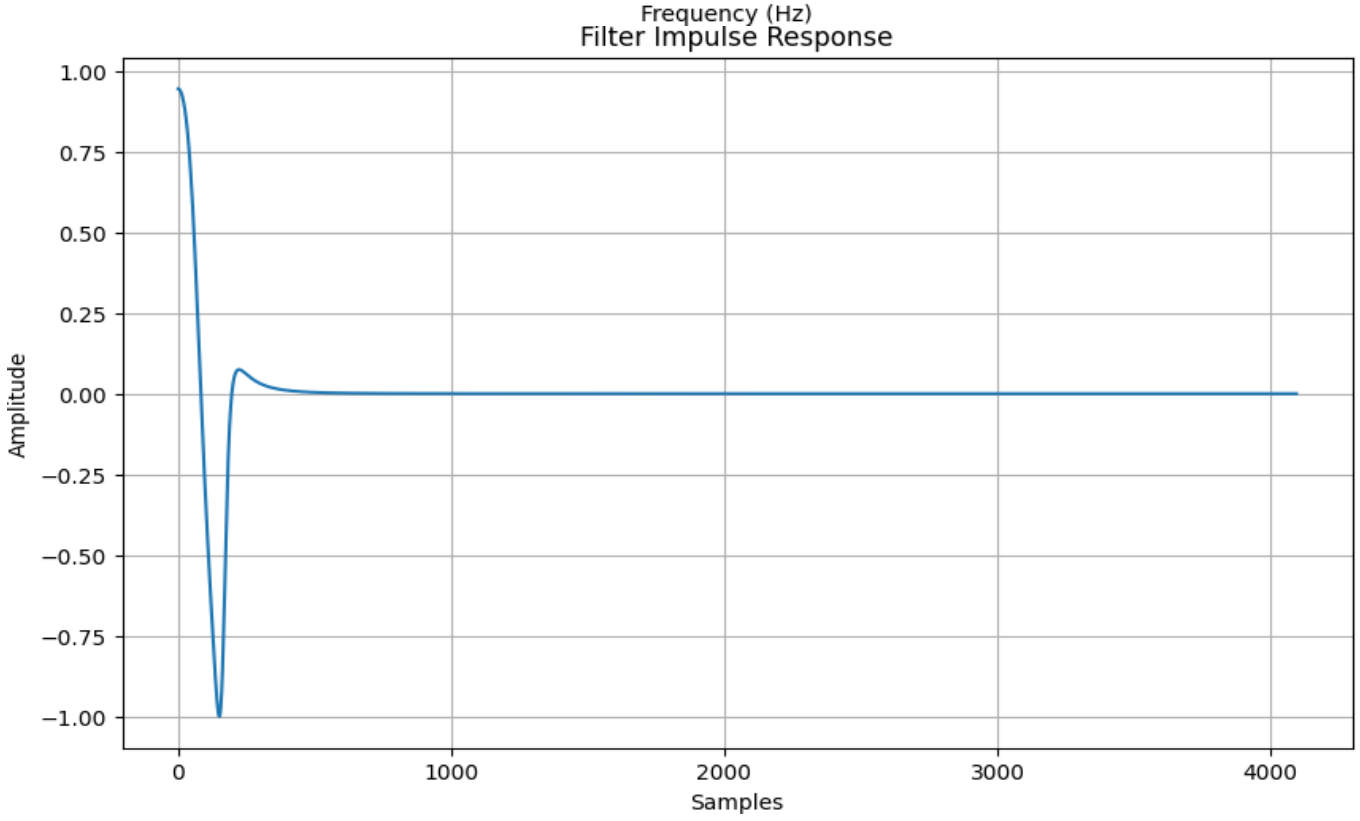


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Que – 2

Generate a discrete sine wave of 1 KHz, 2KHZ, 3 KHZ, 4 KHz, Plot all these

Add all these four signal and plot it in time as well as frequency domain *

Pass through butterworth digital filter having cutff frequency 2 KHZ And plot the output plot in time as well as frequency domain

Pass through chebyschev digital filter having cutoff frequency 3 KHz and plot the output plot in time as well as frequency domain.

Code:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import butter, bilinear, cheby1, lfilter

fs = 16000
duration = 0.01

t = np.arange(0, duration, 1/fs)

f1 = 1000
f2 = 2000
f3 = 3000
f4 = 4000

x1 = np.sin(2 * np.pi * f1 * t)
x2 = np.sin(2 * np.pi * f2 * t)
x3 = np.sin(2 * np.pi * f3 * t)
x4 = np.sin(2 * np.pi * f4 * t)
```

```
plt.figure(figsize=(10, 8))

plt.subplot(2, 2, 1)
plt.plot(t, x1)
plt.title("1 KHz Sine Wave")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid()

plt.subplot(2, 2, 2)
plt.plot(t, x2)
plt.title("2 KHz Sine Wave")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid()

plt.subplot(2, 2, 3)
plt.plot(t, x3)
plt.title("3 KHz Sine Wave")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid()

plt.subplot(2, 2, 4)
plt.plot(t, x4)
```

```
plt.title("4 KHz Sine Wave")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid()

plt.tight_layout()
plt.show()

x = x1 + x2 + x3 + x4

plt.figure(figsize=(10, 5))
plt.plot(t, x)
plt.title("Sum of 1 KHz, 2 KHz, 3 KHz and 4 KHz Signals")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid()
plt.show()

N = len(x)
frequency = np.fft.rfftfreq(N, 1/fs)
X = np.abs(np.fft.rfft(x)) * 2 / N

plt.figure(figsize=(10, 5))
plt.plot(frequency, X)
plt.xlim(0, 5000)
plt.title("Frequency Domain of Combined Signal")
```

```
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.grid()
plt.show()

filter_order = 4
cutoff_butter = 2000

analog_cutoff_butter = 2 * fs * np.tan(np.pi * cutoff_butter / fs)

analog_butter_b, analog_butter_a = butter(
    filter_order,
    analog_cutoff_butter,
    analog=True,
    btype='low'
)

digital_butter_b, digital_butter_a = bilinear(
    analog_butter_b,
    analog_butter_a,
    fs
)

butter_output = lfilter(
    digital_butter_b,
    digital_butter_a,
```

```
    x
)

plt.figure(figsize=(10, 5))
plt.plot(t, butter_output)
plt.title("Butterworth Filter Output - 2 KHz Cutoff")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid()
plt.show()

Y_butter = np.abs(np.fft.rfft(butter_output)) * 2 / N

plt.figure(figsize=(10, 5))
plt.plot(frequency, Y_butter)
plt.xlim(0, 5000)
plt.title("Butterworth Output - Frequency Domain")
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.grid()
plt.show()

cutoff_cheby = 3000
ripple = 0.5

analog_cutoff_cheby = 2 * fs * np.tan(np.pi * cutoff_cheby / fs)
```

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```
analog_cheby_b, analog_cheby_a = cheby1(
    filter_order,
    ripple,
    analog_cutoff_cheby,
    analog=True,
    btype='low'
)

digital_cheby_b, digital_cheby_a = bilinear(
    analog_cheby_b,
    analog_cheby_a,
    fs
)

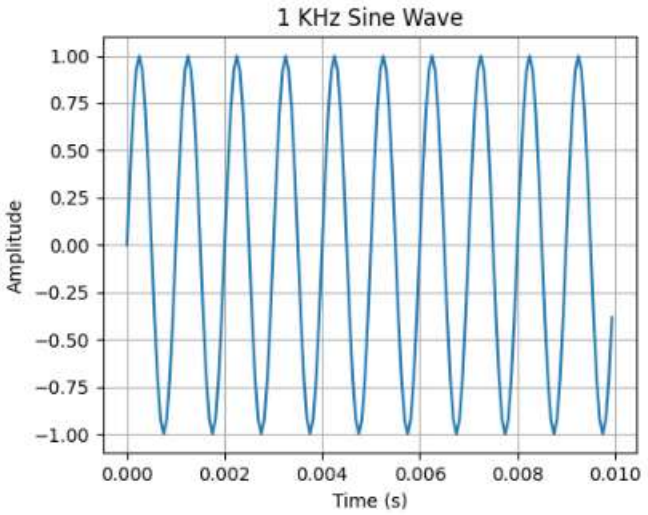
cheby_output = lfilter(
    digital_cheby_b,
    digital_cheby_a,
    x
)

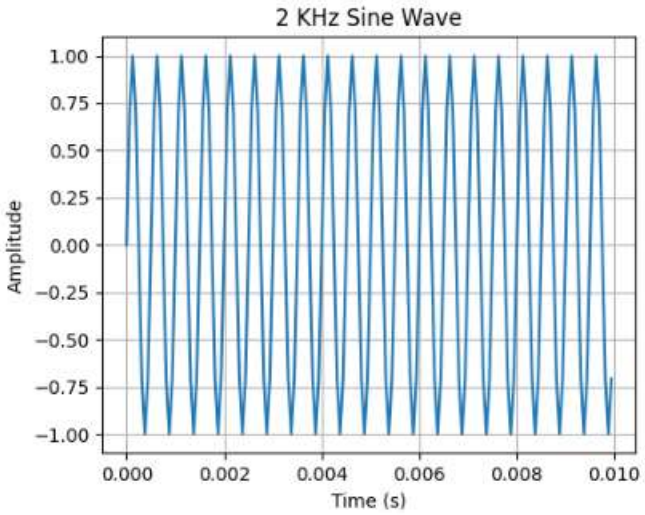
plt.figure(figsize=(10, 5))
plt.plot(t, cheby_output)
plt.title("Chebyshev Filter Output - 3 KHz Cutoff")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
```

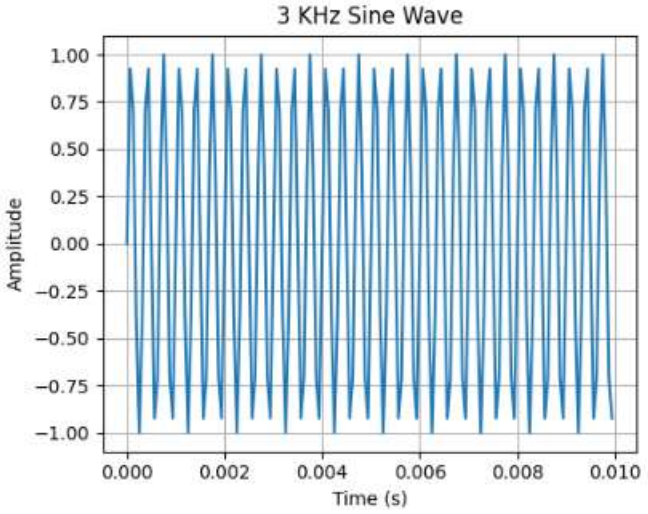
```
plt.grid()
plt.show()
Y_cheby = np.abs(np.fft.rfft(cheby_output)) * 2 / N
plt.figure(figsize=(10, 5))
plt.plot(frequency, Y_cheby)
plt.xlim(0, 5000)
plt.title("Chebyshev Output - Frequency Domain")
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.grid()
plt.show()
```

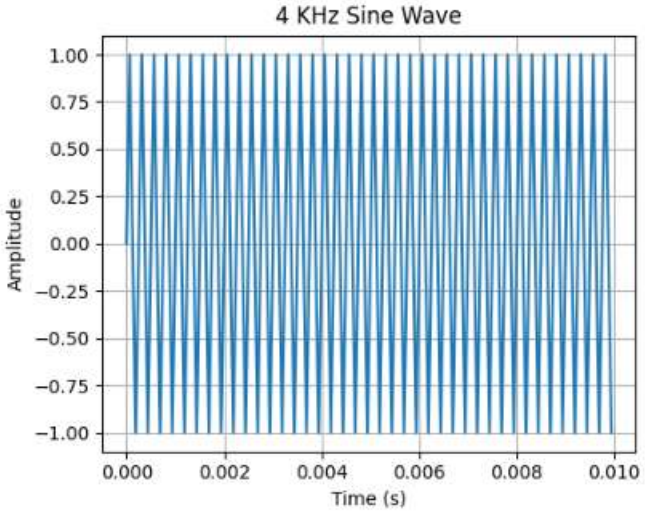
Output:

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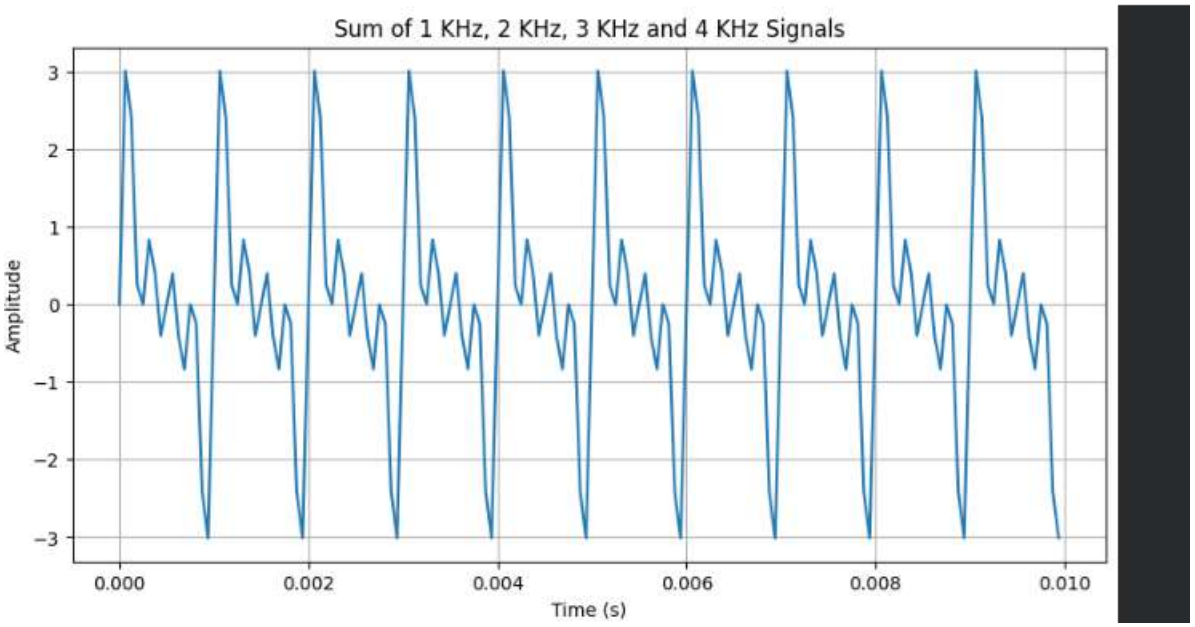


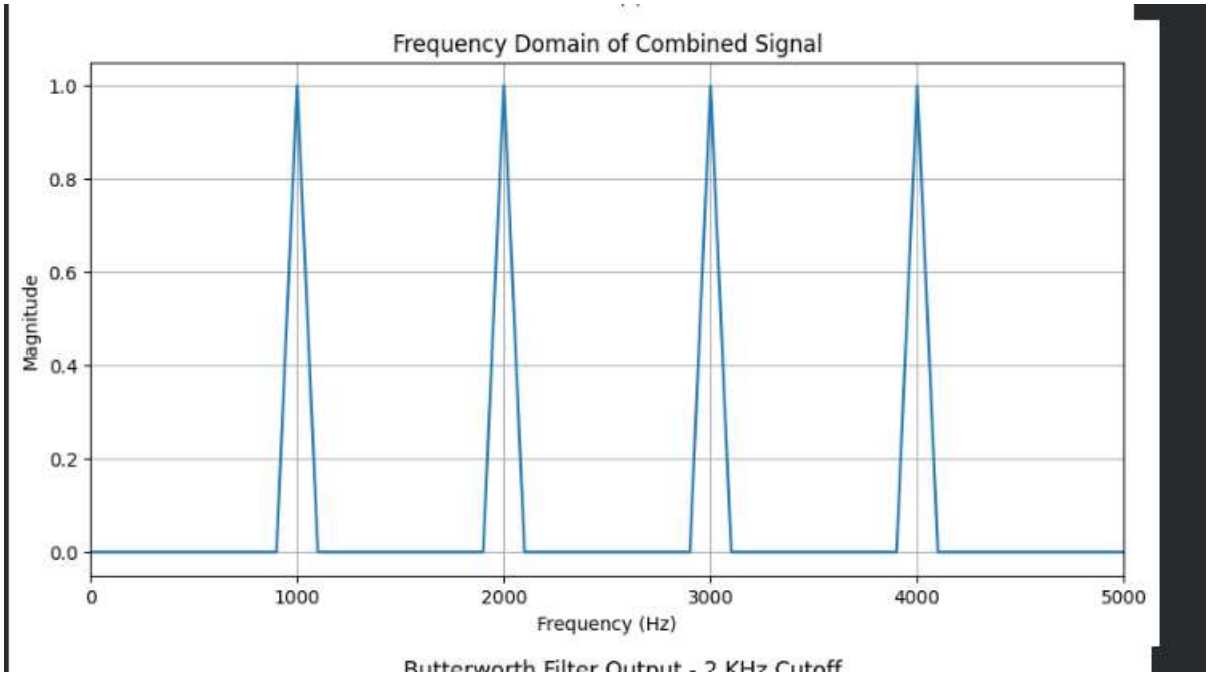






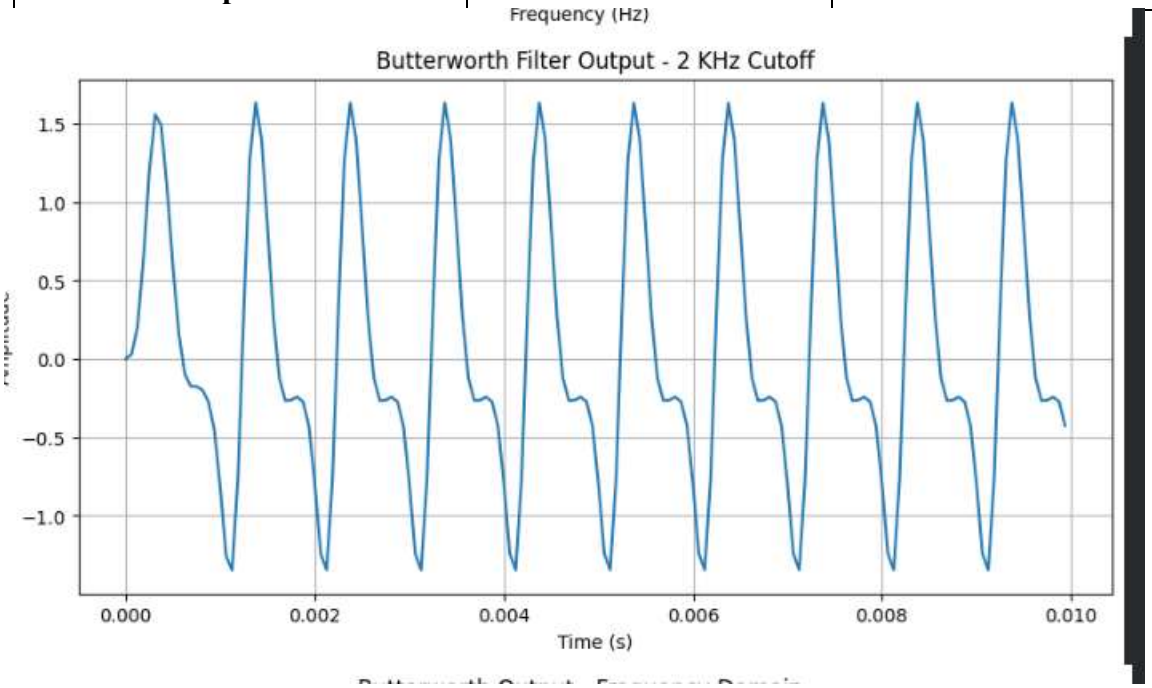
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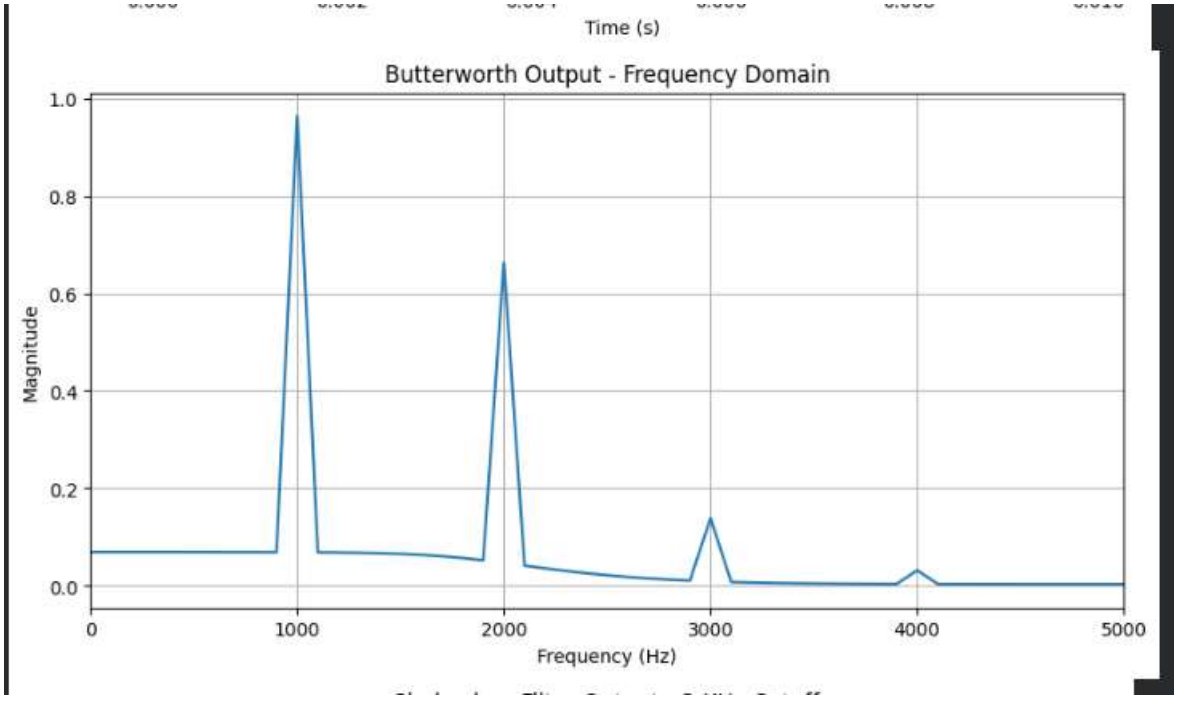




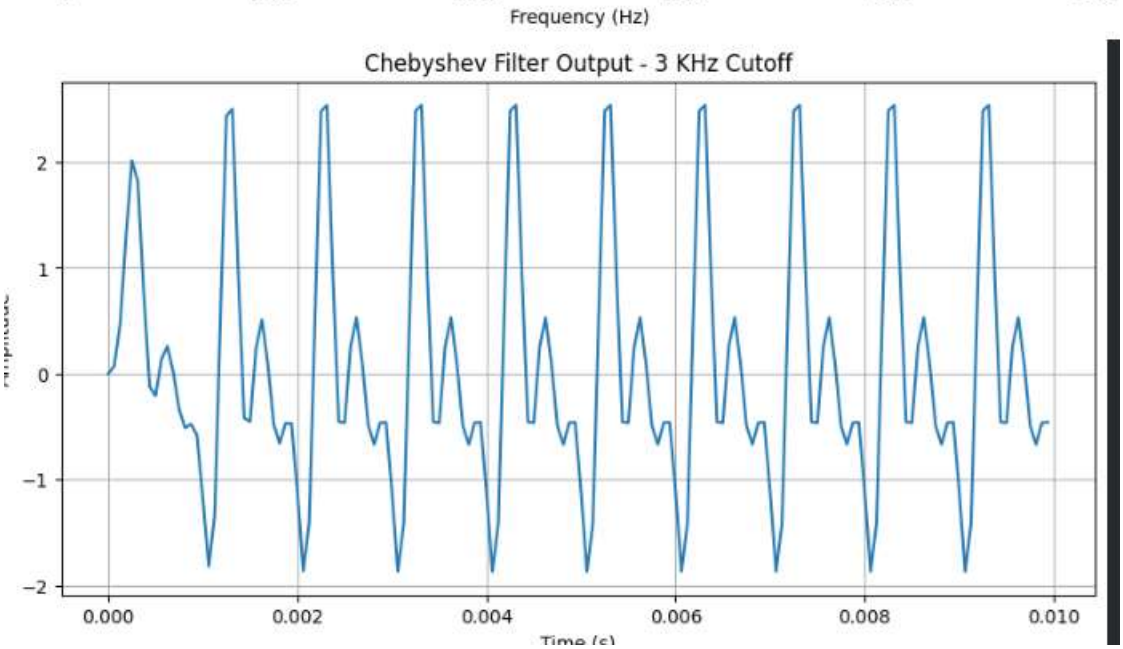
Butterworth Filter Output - 2 KHz Cutoff

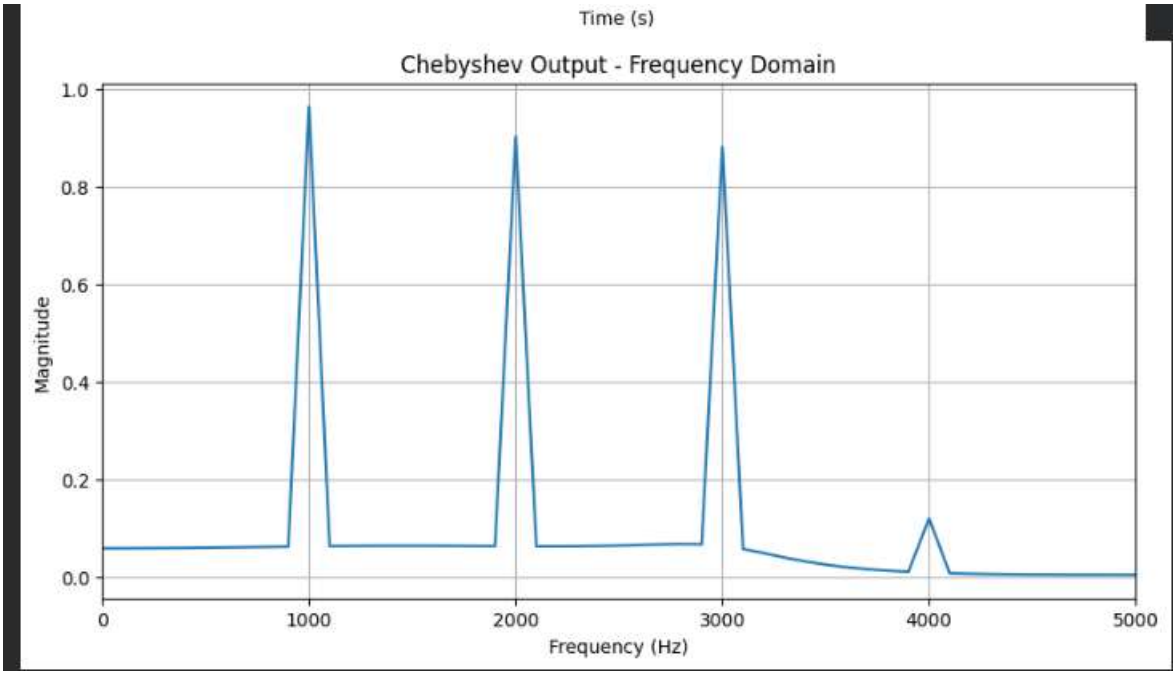
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Conclusion:

In this experiment, four sine waves with different frequencies were generated and added together to form a composite signal. Its frequency components were analyzed using a frequency-domain plot. The composite signal was then processed using Butterworth and Chebyshev digital filters with different cutoff frequencies. The Butterworth filter with a 2 kHz cutoff attenuated the higher-frequency components, while the Chebyshev filter with a 3 kHz cutoff produced a different filtering response. Thus, the characteristics and effects of digital filters on signals were studied and analyzed in both the time and frequency domains.

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Conclusion:

In this experiment, four discrete sine signals of different frequencies were generated and combined into a single signal. The frequency components of the combined signal were observed using the frequency-domain plot.

The combined signal was then filtered using Butterworth and Chebyshev digital filters. The Butterworth filter with a 2 KHz cutoff reduced the higher-frequency components, while the Chebyshev filter with a 3 KHz cutoff allowed more of the lower-frequency components to pass.

From the obtained results, the effect of different digital filters on the signal was successfully studied in both time and frequency domains.